

Translation Circuit Discovery on Qwen Models: Methods, Failures, and the Edge-Pruning Fix

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Abstract

We describe the full pipeline for locating the Spanish-to-English translation circuit in Qwen3-class models (8B & 1.7B), developed iteratively over three methodological failures and two paradigm shifts. Starting from neuron-level activation scans (Phase 1a), we progress through retracted component-level ablation (Phase 1b), a diagnosed capacity failure in differentiable weight masking (DiscoGP / sheaf pruning), and finally a successful edge-pruning protocol that closes the memorization gap and recovers 99% of full-model first-token accuracy with 26% of residual-stream edges. The IOI circuit (Wang et al. [1]) on GPT-2 small serves as the control environment throughout. All code is at github.com/lgemc/mi-lab.

1 Overview

The project proceeds in three phases (§§2–4), each with pre-registered success/failure criteria. The first phase (localisation) went through three full iterations—the initial approach was wrong, the first result was retracted, and the final method is supported by a random-component control. The pipeline is summarised in Figure 1.

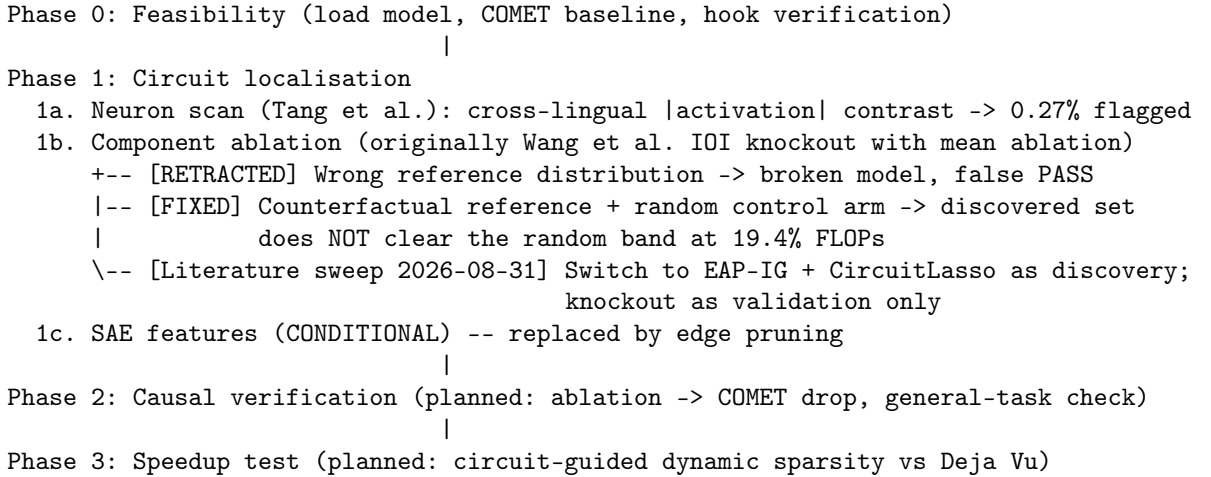


Figure 1: Pipeline overview. Phase 1 consumed most of the effort; Phases 2 and 3 are pending methodological fixes from the Phase 1b failures.

2 Phase 0: Feasibility

Model. Qwen3-8B in bf16 (15.3 GiB CUDA, 36 layers, $d_{\text{model}} = 4096$, 32 heads, $d_{\text{head}} = 128$, 8 KV heads, MLP width 12,288). Loaded via Transformers without 8-bit quantisation—the GB10 had 13 Gi free, which was sufficient. Fast-iteration model: Qwen2.5-3B (bf16, ≈ 6.5 GB). Contrast model: Aya-23-8B.

Baselines. 200 FLORES es→en dev pairs, few-shot WMT-style prompting: **COMET-22** = **0.8712**, **sacreBLEU** = **30.88**, **chrF** = **61.84**. COMET throughput: ≈ 33 s per 100 sentences.

Prompt form. Few-shot WMT-style (3 shots:

Spanish: ... \n English: ...) beats instruction prompting (BLEU 39.3 vs 31.0). The model translates without explicit instruction.

Hook verification. Residual capture and per-head outputs verified at every layer; `decompose()` remainder 0.39% relative (bf16 ULP). Patch round-trips shift logits by exactly 0.0.

3 Phase 1a: Neuron-Level Activation Scan

Protocol from Tang et al. [3] (ACL 2024). For every FFN neuron, compute

$$\text{score}_i = \overline{|a_i|}_{\text{ES}} - \frac{\overline{|a_i|}_{\text{EN}} + \overline{|a_i|}_{\text{control}}}{2}$$

at the MLP down-projection input, over ~ 5 k tokens per condition (6,116 / 5,047 / 5,300). Flag score $> 2\sigma$ ($\sigma = 0.805$).

Result. 1,198 of 442,000 FFN neurons flagged (0.27%). **Concentration is top-only:** 1,185/1,198 in layers 27–35; 85% in layers 32–35; **zero in the bottom quarter**. Peak layer 33 (325 flagged). Outlier: layer 35 neuron 8136 at $\approx 78\sigma$.

Interpretation. Matches Harrasse et al. [4] (language identity in late-layer high-frequency features) and Schüt et al. [5] (output-language decoding in English-shaped representation space).

Contradicts the bottom-layer expectation of Tang et al.—though this is a methodological caveat (see §3.1).

3.1 Caveat

Our statistic is a contrastive mean-absolute-activation difference, not Tang et al.’s LAPE (activation probability + entropy across languages, lowest percentile). The “contradiction” of bottom-layer findings is therefore a misattribution; we measure a different quantity. A raw-Spanish-text control pass would separate “Spanish input neurons” from “output-language / task machinery.”

4 Phase 1b: Component Ablation — Three Iterations

4.1 First attempt (RETRACTED)

Following Wang et al. [1] and Zhang et al. [2], we applied MEAN ABLATION to a candidate set of 9 MLPs in layers 27–35 (19.43% of model MACs) over a WMT-500 dev subset. The ablation means were captured over raw English reference sentences—bare text, no few-shot skeleton, no Spanish:/English: labels, no in-context task.

Result. $\Delta\text{COMET} = 0.643$ against the pre-registered threshold of 0.020 —a “PASS” by a factor of 32. Output collapsed to a a a a a ... (BLEU 6.06).

Three methodological failures (documented in §8.3 of the design doc):

1. **Wrong reference distribution.** Wang et al. [1] §3 require p_{ABC} : preserve the task template, vary only the names. Zhang et al. [2] §3 build X^- on the same rule. Ablating toward raw English removed the model’s ability to continue a prompt at all.
2. **Misattributed method.** Mean ablation is Zhang et al.’s *validation* step (§4.3), not their discovery method (subspace-intervened path patching, §4.1). Reusing the cheap validation step as the expensive discovery step drops the subspace confinement and the random control.
3. **Missing random-component control.** Zhang et al.’s Figure 2 is entirely “Key Heads vs Random Heads, Key MLPs vs Random MLPs.” Without this contrast, “ablating 9 MLPs drops COMET 0.64” is unfalsifiable.

All three failures were latent in the design and were caught only by reading the actual generations rather than the corpus metric. The full analysis is in [wiki/concepts/interpretability/ablation-reference](#).

4.2 Second attempt: Corrected reference + random control

Fix. The counterfactual prompt form preserves every token of the eval prompt but neutralises the translation relation:

```
Text: <es sentence>
Nothing: <en sentence, belonging to a different shot>
... (3 shots) ...
Text: <es sentence>
Nothing:
```

Same skeleton, same shot count, same Spanish source, same shot targets token for token. Language labels neutralised and shot targets rotated off their own sources.

Random control. `scripts/phase1b_random_control.py` draws component sets matched to a target set’s MACs (to within 0.02%), ablates toward the same counterfactual means, and reports the discovered arm against the random band. Each arm carries a degeneracy score (fraction of outputs that are repeated tokens rather than language).

Result. WMT-200 shortlist, baseline BLEU 38.42, degeneracy 0.00:

Scope	Arm	BLEU	Δ BLEU
Model	discovered	0.96	−37.46
Model	random band	0.00–14.80	−23.62– −38.42
Candidate (27–35)	discovered	1.04	−37.38
Candidate (27–35)	random band	0.00–1.96	−36.46– −38.42

The candidate scope (layers 27–35) is the sharper test. Every random draw has the same 7MLP/223head composition; the band is only 1.96 BLEU wide, and the discovered set lands at its **median**. **Separation:** −1.04 BLEU (negative).

Interpretation. At 19.4% of model MACs, all arms are broken models (degeneracy 0.16–1.00). The knockout measures destruction, not localisation. As Kervadec et al. [12] predict, LLM computation is generally dense: there is no 19.4% of the model that can be flattened and leave a language model standing.

4.3 Literature currency check (2026-08-31)

A sweep of 2026 literature after the retraction found three concrete corrections:

1. **Discovery method.** The 2026 survey [15] lists ACDC, EAP, attribution/ALTI ($\sim 100\times$ faster), and CLT attribution graphs. MIB [16] (2504.13151) finds attribution and mask-optimisation methods best for circuit localisation. **Greedy knockout is on neither list.**
2. **SAE premise contested.** MIB’s causal-variable track finds **SAE features are not better than neurons**; supervised DAS wins that track.
3. **Mechanistic ceiling.** Kervadec et al. [12] build a computation-density estimator: LLM processing is **generally dense**; raretokens require higher density; aggressive pruning preserves the output mode while flattening the distribution—a failure mode invisible to corpus MT metrics.

The revised Phase 1b uses **EAP-IG** (attribution) as the primary discovery method, with **CircuitLasso** [10] as a cross-check. Knockout (mean ablation + random control) is kept for validation only. Section 5 reports the implementation and what it reproduces of the sweep it replaces.

5 Attribution as the Discovery Method

The pre-registered replacement is implemented in `src/methods/attribution.py` and checked against the knockout sweep it replaces. Because mean ablation is the study’s intervention, the quantity estimated is that intervention’s first-order term, $(a_{\text{clean}} - a_{\text{mean}}) \cdot \partial m / \partial a$ at the two sites `knockout.ablate` replaces [22, 23]. The cost is one clean pass plus s backward passes for the *whole* lattice, against one generation pass per component for the sweep.

5.1 What the comparison can and cannot say

The sweep is the ground truth and it has a noise floor that has to be stated first: 288 of its 306 rows are single heads scored on half the corpus, with ΔBLEU spread $\sigma = 0.21$ about a mean of -0.04 and only ten of the 288 outside ± 0.5 . A rank correlation over all 306 is therefore mostly a correlation against that spread, and it comes back near zero for every estimator tried ($\rho \in [-0.03, -0.02]$). The answerable comparison is against the 18 components the sweep scored on the full 200 sentences; single-head attributions are summed into their `heads:L` group to match that vocabulary, which first-order attribution licenses because it is linear in the intervention.

5.2 Results (Qwen3-8B, band 27–35, 32 sentences)

Estimator	ρ vs ΔBLEU	top-5 overlap	seconds
Knockout sweep (the measurement)	—	—	7575
EAP, $s = 1$, target	+0.135	0.25	12
EAP-IG path, $s = 5$, target	+0.548	0.67	44
EAP-IG path, $s = 5$, kl	+0.521	0.67	45
EAP-IG path, $s = 5$, target, --against gold	+0.600	0.67	42
EAP-IG path, $s = 5$, kl, --against gold	+0.583	0.67	41

Two results, and the second was not the expected one.

Integration is the whole difference. The first-order term at the clean point is close to useless here ($\rho = +0.135$); integrating the gradient along the path back from the mean-ablated model lifts it to $\rho = +0.548$ and its top-5 overlap from 0.25 to 0.67, for $3.7\times$ the cost and still $172\times$ faster than the sweep. The two estimators agree with each other at only $\rho = +0.100$ (top-20

overlap 0.25), so this is not a refinement: at $s = 1$ and $s = 5$ they are different rankings and only one of them tracks the ablation. This is the saturation failure EAP-IG [24] was written for, and it is the reason the pre-registration named EAP-IG rather than EAP.

The metric choice matters, but only once the two metrics are asking different questions. Teacher-forced on the *unablated model’s own generations*, `target` and `k1` agree at $\rho = +0.940$ (top-20 overlap 0.90) — as they should, since “keep the likelihood of what the full model said” is then a mode-seeking version of “keep the full model’s distribution”. Scoring against the WMT reference instead (`--against gold`) removes that construction and the two separate: $\rho = +0.784$, top-20 overlap 0.67. So the apparent null was an artefact of the target, not evidence against Zhang & Nanda [25], and the reportable form of the result is that on this task the *target* choice dominates the *metric* choice.

Scoring against the reference also tracks the sweep slightly better ($\rho = +0.600$ for `target`, $+0.583$ for `k1`, both at top-5 overlap 0.67), which is what should be expected: ΔBLEU is measured against those same references, so the reference-scored attribution and the measurement it is checked against are asking one question and the self-scored one is asking two.

A caveat that applies to both: what is estimated is a per-token likelihood and what the sweep measures is corpus BLEU over a 64-token greedy generation. Attribution cannot see a component that changes the generation without changing next-token likelihood along the teacher-forced path, and the residual disagreement at $\rho = 0.55$ is an upper bound on how much of that there is, not a measurement of the approximation error alone.

5.3 Node and edge attribution

The same pass runs at edge granularity, scoring (source, reader) pairs rather than components, and a source’s edges sum *exactly* to its node score — two routes to one number, checked to float noise on GPT-2 small (`tests/attribution.py::test_an_edge_attribution_sums_to_the_node_attribution_of_the`). Two things had to be right for that to hold, and both were wrong first. A gradient read by hooking the tensor a destination was handed picks up the residual stream’s other uses as well, counting every downstream path once per upstream destination; the partial derivative through one reader needs a zero tensor added on that reader’s path alone. And every source also writes straight to the unembedding — an edge no ablation-based method enumerates, because nothing ablates it — without which a late layer’s MLP scores near zero per edge while scoring large as a node.

That sum is also what is lost going the other way. Nothing in a node score says which reader the damage was done to, which is the structure post-hoc weight-space methods give up when they collapse back to per-unit scores.

6 Sheaf/DiscoGP Weight Pruning

Parallel to the Phase 1b knockout work, we implemented differentiable weight masking (equivalently, a “sheaf” or DiscoGP-style circuit: a gate per weight over all 28 blocks of Qwen3-1.7B, 1,409,286,144 gates).

6.1 Objective

A learned mask $m \in [0, 1]^{\text{weights}}$ is sampled via the Gumbel-sigmoid trick and applied elementwise to the frozen weights:

$$\mathcal{L} = \underbrace{\text{KL}(p_{\text{masked}} \parallel p_{\text{full}})}_{\text{faithfulness}} + \lambda \underbrace{\|m\|_1}_{\text{sparsity}} + \gamma \underbrace{(\text{accuracy complement})}_{\text{completeness}}$$

6.2 Runs and results

Five runs with different faithfulness objectives and density targets on a word-level translation frame (Spanish: <word>\n English:) with 467 distinct prompts (~ 350 training split):

Run	Density	Held-out	Recovered	Train	Complement	First token
kl-t005	5.2%	0.674	0.34	0.926	0.489	0.016
kl-t005-frame	8.1%	0.624	0.28	0.968	0.527	0.172
gold-t02	15.2%	0.720	0.47	0.988	0.516	0.453
kl-t02-dual	19.1%	0.710	0.45	0.982	0.489	0.297
gold-t02-units	22.8%	0.613	0.26	0.993	0.425	0.383
Chance floor	— — —	0.477	0.00	0.477	— — —	— — —
Full model	100%	— — —	1.00	— — —	— — —	0.629

6.3 Mask capacity failure

Three readings:

- 1. Train/held-out gap is the constant of the sweep.** Train 0.93–0.99 vs held-out 0.61–0.72 at every density, under every objective. This is **1.4 billion free bits fitting ~ 350 examples**.
- 2. The headline is read off the wrong scale.** The metric is answer logit $>$ distractor logit. The floor at all-gates-shut is 0.477. A circuit at 0.720 has recovered 46% of the available range, not 72%.
- 3. The completeness term measures destruction.** Complement accuracy is 0.425–0.527 against a 0.477 floor: noise. McGrath et al. [13] (“Hydra Effect”, 2307.15771) show a healthy complement *should* still work—so the completeness term rewards breaking shared machinery.

Why IOI on GPT-2 small escapes this. Four differences:

	IOI / GPT-2 small	Translation / Qwen3-1.7B
Gates	85M	1,409M
Distinct prompts	templated, 10k+	467
Contrast	logit diff within one prompt	answer vs another’s answer
Answer = argmax	98.4%	52.4%

The within-prompt contrast cancels “how do I continue text at all.” The gate/prompt ratio differs by ~ 3 orders of magnitude.

7 Edge Pruning: The Fix

Following Bhaskar et al. [14] (“Finding Transformer Circuits with Edge Pruning”, arXiv:2406.16778), we prune **edges of the residual-stream computation graph** rather than weights.

Capacity argument. A 28-layer, 16-head model admits ~ 14 k edges against 1.4B weights—five orders of magnitude less freedom. An edge is also the unit the word “circuit” actually refers to.

7.1 GPT-2 small IOI: Validation

2028 edge gates, ~380 distinct prompts, KL faithfulness, 2500 steps.

	Weight mask	Edge mask
Gates	85M	2,028
Train	0.93 – -0.99	0.712
Held-out	0.61 – -0.72	0.685
Gap	+0.27	+0.027

The memorisation gap closes from +0.27 to ± 0.03 .

Density/accuracy relation is monotone:

Edges open	100%	64.7%	45.7%	32.0%	26.2%	21.5%	17.3%
7.6%							
Held-out	1.000	1.000	1.000	1.000	0.969	0.938	0.750
0.685							

Qualitative match to Wang et al. At 7.6% density: all three duplicate-token heads (0.1, 0.10, 3.0), induction head (5.9), two S-inhibition heads (7.3, 7.9), two of three name movers (9.6, 9.9), and the specific edges `head:7:9 → attn:9`, `head:7:3 → attn:10`, `head:7:9 → attn:10` (S-inhibition writing into name movers’ input).

Random-edge control. Five random 155-edge subsets:

	Held-out	First token
Learned, 155 of 2028	0.685	0.472
Random (5 seeds)	0.370 – -0.480	0.000(every seed)

7.2 Qwen3-1.7B translation: Main result

14,140 edge gates (vs 1.4B weight gates), 900 steps, 25% density target, the same word-level translation frame and prompt pool as §6.

	Weight mask (best)	Edge mask
Gates	1,409,286,144	14,140
Density	15.2% of weights	26.3% of edges (3,720)
Held-out ranking	0.720	1.000
Recovered	0.47	1.00
Train – held-out	+0.27	0.000
First token	0.453 (64% of full)	0.624 (99.2% of full)
Complement	0.516	0.462
Wall clock	1h58m	42m

Random-edge control (3 seeds).

	Held-out	First token
Learned, 3,720 of 14,140	1.000	0.624
Random (3 seeds)	0.446/0.478/0.462	0.000(every seed)

Generations (greedy, held-out prompts, 6 tokens):

Spanish:	hoy \n	English:	full: today, circuit: today
Spanish:	pantalla \n	English:	full: screen, circuit: screen
Spanish:	longitud \n	English:	full: length, circuit: length
Spanish:	compartir \n	English:	full: share, circuit: share
Spanish:	paranoia \n	English:	full: paranoia, circuit: paranoia

Full translations, not degenerate repetition. The continuation after the answer is still degraded (multi-position KL is the next improvement).

8 Summary of Methods and Tools

- **Models:** Qwen3-8B (primary, bf16), Qwen2.5-3B (fast iteration, bf16), Qwen3-1.7B (sheaf runs), GPT-2 small (IOI control).
- **Frameworks:** TransformerLens hooks (Phase 0/1a/1b), nnsight (future), PyTorch + uv, Hydra composition, MLflow tracing.
- **Metrics:** COMET-22 (XLM-R-large), sacreBLEU, chrF, KL divergence, logit difference, degeneracy, first-token accuracy, recovered = (accuracy - chance) / (full - chance).
- **Data:** WMT14 ES→EN (30k dev, 3k test), FLORES-100 es_dev (1k sentences), word-level translation frame (467 prompts).
- **Methods employed:**
 - Neuron-level cross-lingual activation scan (Phase 1a).
 - Mean ablation knockout with counterfactual reference + random control (Phase 1b, first two iterations – now retired as a discovery method).
 - Attribution / EAP-IG (Phase 1b, revised – primary discovery, implemented in `src/methods/attribution` see §5). CircuitLasso as a cross-check remains unimplemented.
 - Differentiable weight masking / DiscoGP sheaf (independent parallel track – identified the capacity failure).
 - Edge pruning (Bhaskar et al. 2024) – the fix that closes the gap and produces the only unretracted result in this document.
 - ArXiv-supported literature methods: subspace-intervened path patching (Zhang et al. 2025), ACDC (Conmy et al.), EAP-IG, CircuitLasso, DAS, transcoders.

9 Pre-registered Decision Criteria

Circuit exists	Phase 1: component set $\leq 25\%$ FLOPs whose ablation drops $\Delta\text{COMET} \geq 2.0$ pts
Circuit is separable	Phase 2b: general-task ΔCOMET from same ablation $<$ half of translation drop
Speedup claim	Phase 3: Arm C beats Arm B by ≥ 0.5 COMET at the same FLOPs budget
Negative	Arm C $\approx$ Arm B at every budget — publish the localisation + negative efficiency res

Amendment (2026-08-31): No ΔBLEU / ΔCOMET is quotable from an arm whose degeneracy is non-zero. Survival is a precondition of the test, not an outcome of it.

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